
EE 608: Digital Image Processing

Course Code	EE 608
Course Name	Digital Image Processing
L-T-P-C	3-0-2-4
Prerequisites	Basics of signal processing and Probability theory
Intended for	UG/MS/PhD
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Introduction to digital image processing:** What is image processing, Different types of images, Visual perception, Image sensing and Acquisition, Quantization, Sampling, color image processing, Revision of Mathematical concepts for image processing. [3 Lectures]
- **Intensity transformation, Filtering in spatial and Frequency domain:** Image negatives, Log transformations, Histogram processing, Spatial filter: smoothing and Sharpening, Discrete Fourier transform, properties of 2-D DFT, Image smoothing and Sharpening in Fourier domain. [8 Lectures]
- **Image transforms:** Two-dimensional orthogonal and Unitary transforms, Optimum transform, Properties of Unitary transforms, 2D DFT, Cosine transforms, Hadamard transforms, KL transforms, Comparison of image transforms. [5 Lectures]
- **Edge detection:** Gradient and Laplacian based edge detection, Diffusion based edge detection: Isotropic and anisotropic diffusion. [3 Lectures]
- **Wavelet transform for Image Processing:** Multi resolution expansion, Wavelet functions, Wavelet Series expansion, Continuous and Discrete Wavelet transforms, Wavelet transforms for two-dimensional signals (images), Applications of wavelet transforms for edge extraction, noise suppression. [5 Lectures]
- **Image segmentation:** Thresholding, region-based Morphological Watersheds, Bayesian based image segmentation. [5 Lectures]
- **Image restoration and reconstruction:** Models of image degradation, noise models, Spatial and Frequency domain based approaches for image restoration, Inverse filtering, Wiener Filtering, Bayesian de-noising. [5 Lectures]
- **Image Compression:** Spatial and Temporal redundancy, Basic image compression models, compression standards, basic compression methods: Huffman coding, Run-length coding, Block transform coding, Predictive coding. [4 Lectures]
- **Color Image Processing:** Color Fundamentals, Color Models, Color transformation, smoothing, sharpening and edge detection in color images. [4 Lectures]

Text Books

1. R. C. Gonzalez and R. E. Woods, Digital Image Processing, 3rd Edition, Pearson Education, 2009
2. Anil K Jain, **Fundamental of Digital Image Processing**, Prentice Hall, 1989

References

1. A. C. Bovik, **The essential guide to image processing**, 2nd Edition, Academic Press, 2009
2. A. M. Teckalp, **Digital Video Processing**, Prentice Hall, 1995.